Lab Quick Start: Introduction to AI and Python

BMED365-2027: Computational Imaging, Modeling and AI in Biomedicine

BMED365

Spring 2027

Overview

1. Introduction to the Course
2. Python for Medical Professionals
3. Jupyter Notebooks
4. Important Libraries
5. Notebook Walkthrough

Welcome to BMED365

Course objectives:

- Understand fundamental AI concepts.
- Apply machine learning to medical data.
- Get familiar with Python as a tool.
- Reflect on ethics and trustworthy AI.

Structure:

- **Lab 0:** ML with PyCaret (AutoML).
- **Lab 1:** Network analysis (PSN).
- **Lab 2:** Deep Learning (Images/ECG).
- **Lab 3:** Generative AI (LLM).
- **Lab 4:** Computational Imaging.
- **Lab 5:** Computational Modeling.
- **Projects:** Team + Individual.

Why Python?

Python has become the de-facto standard in AI and Data Science.

- **Readability:** Syntax resembles English pseudo-code.
- **Ecosystem:** Enormous libraries for everything from statistics to image analysis.
- **Community:** Great support and many examples online (Stack Overflow, ChatGPT).

"Python is executable pseudocode."

Prompt

"A friendly, stylized python snake coiling around a medical cross, 3d render, cute, clean background, high quality, medical blue and white color scheme."



Python vs. Other Languages

Language	Advantages	Disadvantages
Python	Simple, AI standard, flexible	Slower than C++, GIL
R	Excellent for statistics	Less support for Deep Learning
MATLAB	Industry standard in engineering	Expensive, proprietary
C++	Extremely fast	High learning curve, complex

In BMED365, we use Python because the libraries (PyTorch, Scikit-learn) are best developed here.

The Jupyter Notebook Environment

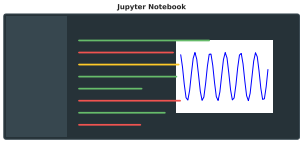
A **Notebook** (.ipynb) combines:

1. **Executable code** (Python cells).
2. **Rich text** (Markdown, LaTeX formulas).
3. **Visualizations** (Plots directly in the document).

This promotes *reproducible research*.

Prompt

"Futuristic coding interface with visualization of medical data, holographic, blue and white theme, Python code snippets floating in the air, clean and modern."



Markdown and LaTeX in Jupyter

We can write formulas directly in text cells using $:

Example:

```
The mean value is given by $\mu = \frac{1}{n} \sum x_i$
```

Result: The mean value is given by $\mu = \frac{1}{n} \sum x_i$

This is useful for documenting mathematical methods in lab reports.

NumPy: Numerical Python

The foundation for all scientific computing in Python.

- Introduces `ndarray` (n-dimensional arrays).
- Much faster than regular Python lists due to C implementation and vectorization.

Mathematical operation

If A and B are matrices, `A * B` in NumPy is typically element-wise multiplication, while `A @ B` is matrix multiplication.

Pandas: Data Processing

The main tool for tabular data (like Excel, but more powerful).

```python
import pandas as pd

# Read data
df = pd.read_csv('patient_data.csv')

# Inspect first rows
print(df.head())

# Statistics
print(df.describe())
```

Important concepts:

- `DataFrame`: The entire table.
- `Series`: A column.
- `.loc[]`: Get data by name.
- `.iloc[]`: Get data by index (position).

Matplotlib & Seaborn: Visualization

- **Matplotlib:** The "grandfather" of plotting. Powerful, but can be verbose.
- **Seaborn:** Built on Matplotlib. More attractive default design, simpler for statistical plots.

A good visualization can replace a thousand numbers.

We will use these to visualize confusion matrices, training curves, and feature importance.

Overview of ‘quickstart-ai-python.ipynb‘

The notebook is divided into the following sections:

1. **Python Basics:** Variables, lists, loops, functions.
2. **NumPy:** Vectors and matrices.
3. **Pandas:** Load and explore a "Heart Disease" dataset.
4. **Visualization:** Plotting histograms and correlations.
5. **Scikit-learn intro:** A simple linear regression model.

Tips for Lab Work

- **Run cells in order!** Variables must be defined before they are used.
- **Tab completion:** Press `Tab` to see available functions.
- **Help:** Use `?` after a function for documentation.

```
pd.read_csv?
```

- **Restart Kernel:** If things get stuck, restart the kernel (Kernel → Restart).

The Road Ahead

When you are finished with the quick start, you are ready for **Lab 0**, where we will begin with real machine learning using PyCaret.

Good luck with coding!